

Vat Photopolymerization of Liquid Metal Nanoparticle-Integrated Hydrogels

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Hydrogels are water-rich, cross-linked polymers widely applied in biomedicine, tissue engineering, wound healing, and 3D bioprinting due to their biocompatibility and structural versatility. Additive manufacturing, particularly vat photopolymerization (VPP) methods such as stereolithography (SLA) and digital light processing (DLP), has enabled high-resolution fabrication of complex hydrogel architectures. However, the precision and performance of VPP-printed hydrogels remain constrained by light scattering, uncontrolled polymerization kinetics, and the poor solubility or toxicity of conventional photo-absorbers. Here, polymer-coated liquid metal nanoparticles (pLMNPs) are introduced as multifunctional additives that address these challenges. Prepared using reversible addition–fragmentation chain-transfer (RAFT) polymer coatings in aqueous solution, pLMNPs exhibit excellent colloidal stability and are uniformly distributed within the hydrogel resins. Their incorporation improves light focusing, reduces scattering, and regulates polymerization kinetics, enabling enhanced print fidelity. Through dynamic covalent bonding between Ga and abundant carboxylic acids/amines on gelatin methacryloyl (GelMA), LMNPs reinforce hydrogel networks, nearly doubling both Young's modulus and compressive toughness. Beyond structural enhancement, pLMNPs impart photothermal responsiveness and antibacterial activity while remaining stable in aqueous media. This work establishes a versatile nanoparticle-assisted strategy for VPP-based 3D printing of hydrogels, offering new opportunities for the design of high-resolution, mechanically robust, and multifunctional hydrogels for biomedical implants and wearable devices.

1. Introduction

Hydrogels, as 3D, cross-linked polymer networks with high water content, are widely used in biomedicine, tissue engineering, wound healing, and 3D bioprinting because of their biocompatibility, extracellular-matrix-like structure, porous nature, and excellent cell-supporting properties.^[1–6] Traditionally, hydrogels have been fabricated onto 2D or simple 3D structures using molding, soft lithography, or photolithography, which requires elaborate fabrication processes or costly apparatus.^[7,8] The emergence of additive manufacturing (3D printing) has so far critically impacted hydrogel fabrication by enabling the creation of customizable, complex geometries with high structural resolution—often below 100 μm in high-end systems.^[8–10] Meanwhile, many 3D printing techniques allow the chance of avoiding de-casting and other post-defects during the handling of hydrogels.

Vat photopolymerization (VPP) is an additive manufacturing process in which a vat of liquid photopolymer resin is selectively cured layer by layer using a light source (e.g., a laser or projector). As one of the earliest 3D printing methods, VPP continues to play a key role in advancing

high-resolution prototyping and customized manufacturing.^[11] So far, various light-assisted VPP techniques have been used for the fabrication of hydrogels,^[12] such as stereolithography (SLA) and digital light processing (DLP) 3D printing methods.^[13–15] Comparing to the nozzle-based 3D printing techniques, VPP have shown to offer better spatial resolution, finer details, enhanced pattern fidelity, and faster fabrication speeds, owing to their unique pixel-based XY resolution.^[15] Unlike many traditional 3D printing methods, VPP imposes fewer viscosity constraints on printable materials, enabling the use of a broad range of monomers, including (meth)acrylates, (meth)acrylamides, vinyl ethers, and epoxides, which can support the versatile design of hydrogels.^[16]

One of the key challenges for successful VPP-based printing is formulating resins with high photoreactivity and appropriate photoinitiators (e.g., lithium phenyl-2,4,6-trimethylbenzoylphosphine abbreviated as LAP) to achieve

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The ORCID identification number(s) for the author(s) of this article can be found under <https://doi.org/10.1002/adfm.202523767>

DOI: 10.1002/adfm.202523767

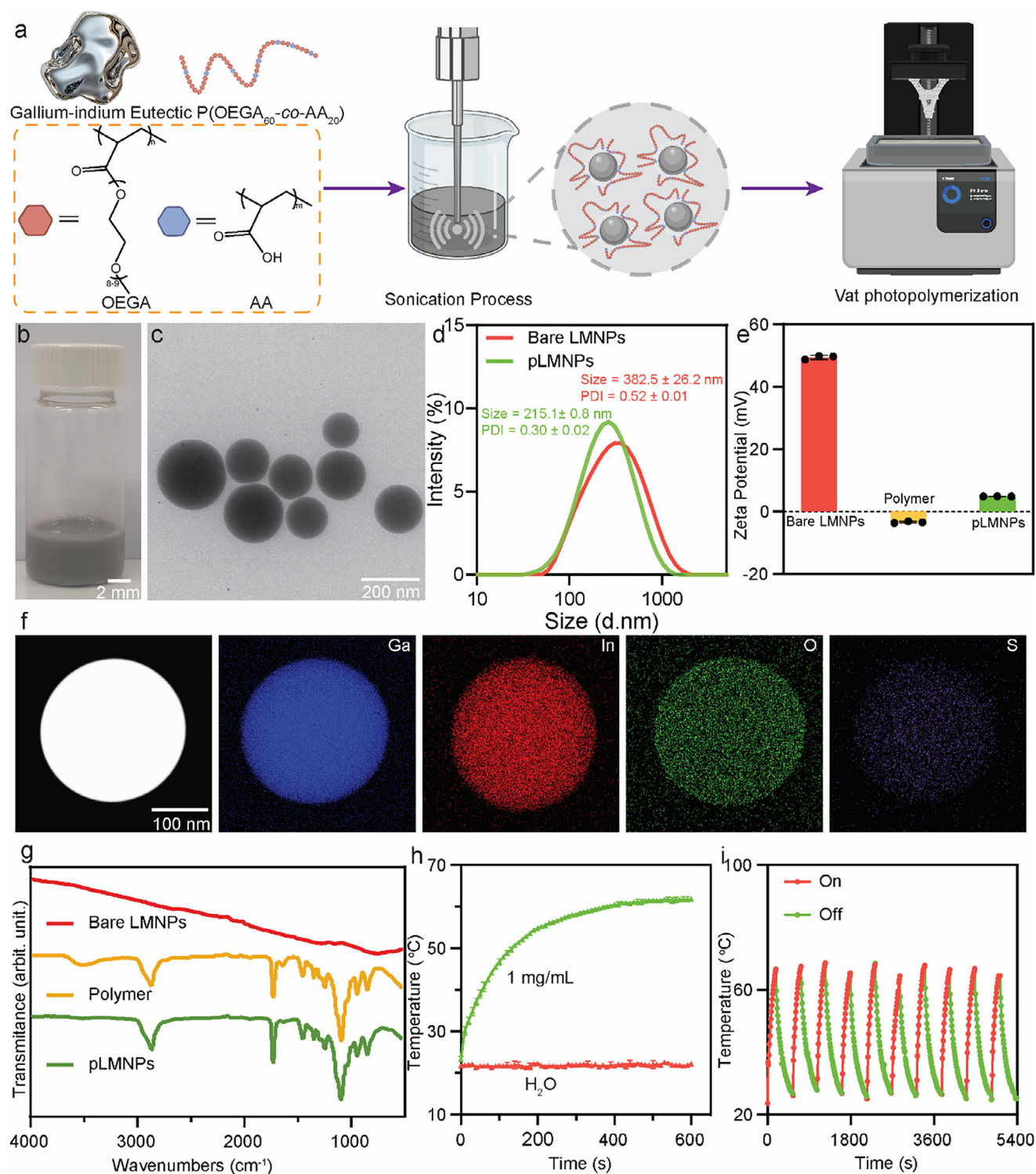


Figure 1. Production and characterization of pLMNPs particles. a) Schematic illustration of the pLMNPs preparation and vat photopolymerization; b) Photograph of pLMNPs solution; c) High contrast TEM image of pLMNPs (gap between the droplets defines the coating thickness – the coating is not seen directly due to the high contrast); d) Size distribution of pLMNPs and bare LMNPs using DLS, and e) Zeta potential of pLMNPs, polymer, and bare LMNPs; f) HADDF and EDS mappings of a pLMNPs droplet; the detected elements include Ga, In, O and S; g) FT-IR spectra of pLMNPs, polymer, and bare LMNPs; h) Photothermal effects of 1 mg mL⁻¹ pLMNPs and water under 1 W cm⁻² 808 nm NIR laser; i) Photothermal stability during ten heating and cooling cycles.

efficient photopolymerization and precise curing.^[17] However, very often, the lack of control over the rapid polymerization kinetics lead to over-crosslinking, which can produce undesired shapes and defects within the hydrogels.^[18] Altogether as a whole, achieving high-resolution 3D-printed hydrogels using 3D printing remains challenging because of ultraviolet (UV) light scattering during the process that result in curing regions beyond the intended exposure areas in high water-containing resins, blurring fine details.^[19] Although photo-absorbers such as tartrazine, curcumin, anthocyanin, quinoline yellow, and carmine can potentially adjust the light exposure conditions, the low light absorption efficacy and limited solubility in hydrogels hinders their usage in VPP.^[17]

In recent years, the incorporation of nanoparticles (NPs) into VPP resins has emerged as an effective strategy to manipulate light propagation, improve mechanical reinforcement, and introduce new functionalities.^[20–22] For example, metal nanoparticles, silver NPs, gold NPs, copper NPs and etc., have been reported to act as optical absorbers for regulating light penetration and spatial resolution to synergistically improve printing resolution and fidelity in both the X-Y plane and the Z direction.^[23,24] Nevertheless, a major limitation persists: nanoparticle aggregation and sedimentation often lead to phase separation and non-uniform light fields in the printing resin, deteriorating both printability and final structural integrity. Furthermore, most existing studies have focused on rigid or conductive polymer composites rather than soft hydrogel systems, leaving a significant research gap in developing nanoparticle-stabilized hydrogel resins that achieve both optical control and mechanical tunability.^[25,26] We hypothesize that devising novel 3D printing hydrogel resin systems, with sufficient stability for NPs and/or exploiting the production of the NPs during printing,^[27,28] can synergistically augment printing control and fidelity, as well as the mechanical properties of the resulting composites or even additional functionalities.

Previously, liquid metal nanoparticles (LMNPs) with high surface reactivity were reported to be coordinately stabilized by reversible addition–fragmentation chain-transfer (RAFT) polymers with different ligands such as thiols, trithiolcarbonate, carboxylic acids, and diphosphoric acids, thereby enabling high dispersity of LMNPs in various aqueous solvents.^[29–33] Herein very differently, a novel printing strategy for fabricating hydrogels via VPP is provided by introducing pLMNPs as both a photo-absorber and a nanofiller. LMNPs have demonstrated strong photo-absorbing properties and have been employed in photonanomedicine.^[34–36] pLMNPs exhibit high colloidal stability in three representative 3D printing hydrogel resins, including gelatin methacryloyl (GelMA), N-isopropylacrylamide (NIPAM), and 2-hydroxyethyl acrylate (HEA)/poly(vinyl alcohol) (PVA) resins. Furthermore, the fabrication of pLMNP-integrated hydrogel through DLP printing technology is explored to achieve intricate geometries. The incorporation of pLMNPs into the printing resin effectively reduces light scattering, enhances light focusing onto transparent resins, and moderates polymerization kinetics, resulting in high-resolution 3D-printed hydrogel structures. This is due to the metallic properties of LMNPs and their favorable photo-absorption. Additionally, pLMNPs contribute to enhanced mechanical properties by forming dynamic cross-linked networks with polymer chains through coordination

bonding, reinforcing the structural integrity of the printed hydrogels. Examples of 3D printing are presented, and the 3D-printed hydrogels are demonstrated to possess potential for antibacterial applications owing to the intrinsic photothermal properties of pLMNPs.

2. Results

2.1. Preparation and Characterization of pLMNPs in Water for VPP Printing

Generally, nanoparticle-integrated VPP resins are prepared by directly dispersing nanoparticles into a mixture of monomers, photoinitiator, and cross-linker. For example, we previously formulated LMNP-integrated resins in an organic solvent via direct sonication,^[30] printed by VPP, these resins enabled shape-memory polymer soft-robotic components.^[37] However, when LMNPs are mixed with aqueous hydrogel monomers, their highly reactive interfaces can trigger unintended polymerization during sonication—via surface oxidation, hydrolysis, charge accumulation, and radical generation—prematurely increasing resin viscosity and degrading print fidelity.^[38] This premature polymerization can lead to increased resin viscosity, which ultimately hinders the high-quality 3D printing process.

To address this premature polymerization challenge, we propose a two-step strategy for 3D printing LMNP-integrated hydrogels. This involves first preparing polymer-coated LMNPs in low-temperature aqueous solution, followed by blending them with the hydrogel resin for the 3D printing process (Figure 1a). This two-step process takes advantage of initial polymer coating, which suppresses continuous interfacial interactions of LMNPs with precursors, with low-temperature conditions that mitigate LMNP excessive surface oxidation—both factors contributing to the formation of uniform LMNPs,^[38] and resin of acceptable viscosity.

To generate highly stable LMNPs, we propose a direct sonication approach in ice water bath in which bulk EGaIn is dispersed in an aqueous solution containing the polymers. First, a block copolymer with carboxylic acid groups was prepared to increase the stability of LMNPs. In this regard, block copolymer poly(oligo(ethylene glycol) methyl ether acrylate₆₀-co-acrylic acid₂₀) (P(OEGA₆₀-co-AA₂₀)) was synthesized using RAFT polymerization.^[29] ¹H NMR spectra (Figure S1, Supporting Information) of the P(OEGA₆₀-co-AA₂₀) confirmed the successful synthesis. The as-prepared block copolymer was dissolved in Milli-Q water and mixed with bulk EGaIn, followed by sonication for 20, 30, and 60 min, respectively. The resulting dispersions exhibited average hydrodynamic diameters of 253.0 ± 17.7, 210.9 ± 9.8, and 220.0 ± 12.1 nm (Figure S2a, Supporting Information), indicating that sonication beyond 30 min yielded pLMNPs of similar size. Therefore, a 30 min sonication time was chosen to obtain monodispersed polymer-coated LMNPs (pLMNPs) of a deep gray color in solution as identified in photography (Figure 1b) and transmission electron microscopy (TEM) imaging (Figure 1c).

The morphology and size distribution of pLMNPs were characterized by TEM and dynamic light scattering (DLS). The pLMNPs possessed spherical morphology of diameters ≈ 158.6 ± 26.2

nm as also confirmed by TEM imaging (an example presented in Figure 1c). As per Figure 1d, pLMNPs display a mono-peak size distribution with a hydrodynamic size of 215.1 ± 0.8 nm, which is consistent with the TEM analysis. Notably, pLMNPs possess better dispersity and smaller size as demonstrated by the single peak in the hydrodynamic size of 215.1 ± 0.8 nm and the polydispersity index (PDI) of 0.31 ± 0.02 as compared with that of bare LMNPs with size of 382.5 ± 26.2 nm and PDI of 0.54 ± 0.01 , in which the polymer coating could largely reduce the aggregation of LMNPs through forming a passivation layer onto the surface of LMNPs. The reduced size of pLMNPs in comparison to LMNPs is in agreement with the previously reported effect of added polymer to the solution during sonication.^[39,40] Moreover, the substantial change in surface charge from 49.40 ± 0.64 to 4.86 ± 0.06 mV further demonstrated the successful coating by P(OEGA₆₀-co-AA₂₀) (Figure 1e). The nearly neutral surface charge of pLMNPs indicates that their stabilization also primarily arises from the steric repulsion provided by the polymer coating, rather than electrostatic interactions.^[41] We will show that its steric stabilization plays a crucial role in enabling the formation of a stable LMNP-integrated resin suitable for the 3D printing process.

We further confirmed the polymer coating of LMNPs by using TEM energy-dispersive X-ray spectroscopy (EDS) mapping and spectrum. As shown in Figure 1f and Figure S2b (Supporting Information), the co-localization of Gallium (Ga), Indium (In), Oxygen (O), and Thiol (S) signals indicates the successful surface coating of polymers onto LMNPs. No significant surface oxidation of Ga and dealloying phenomena were observed from the TEM-EDS images and spectrum, consistent with the low average O content of ≈ 0.87 wt.% (Figure S2b, Supporting Information). The surface coating of LMNPs was further confirmed by the O–H stretching vibration (≈ 3510 cm⁻¹) observed in the FT-IR spectrum of the pLMNPs complex, indicative of Ga–COOH coordination bonds formed between the LMNPs and the carboxylic acid-terminated polymer (Figure 1g). This strong coordination interaction is critical for stabilizing the nanoparticles in aqueous resin formulations. Additionally, the rest of the FT-IR spectrum of pLMNPs displays characteristic peaks corresponding to the coated polymer, further supporting successful surface functionalization.

The photothermal properties of pLMNPs were evaluated under 808 nm near-infrared (NIR) laser irradiation at a power intensity of 1 W cm^{-2} .^[42,43] As shown in Figure 1h, the temperature of the pLMNPs solution rapidly increased to 50°C within 2 min, reaching an equilibrium temperature of $61.8 \pm 0.6^\circ\text{C}$ after 10 min. Increasing the power intensity to 2 W cm^{-2} resulted in a peak temperature exceeding 75°C (Figure S3a, Supporting Information). As shown in Figure S3b (Supporting Information), the temperature rise exhibited a clear concentration dependence, with LMNP concentrations ranging from 0.2 to 2 mg mL^{-1} . Higher concentrations and power densities led to faster heating and shorter times to reach equilibrium. The photothermal stability of pLMNPs was further assessed through multiple heating-cooling cycles (3 min on and 6 min off). Consistent heating and cooling profiles were observed over ten photothermal cycles (Figure 1i), demonstrating the robust photothermal stability of the pLMNPs system.

2.2. Printability of pLMNPs-Integrated Hydrogel Resins

VPP-based printing relies on the selective curing of liquid photopolymer resin; therefore, achieving uniform, defect-free printing requires hydrogel resins with excellent colloidal stability, favorable rheological properties, and robust photopolymerization performance, especially when introducing nanofillers.^[44,45] However, unlike conventional inorganic nanoparticles, LMNPs possess high cohesive energy and a fluidic nature, making them prone to spontaneous coalescence. This behavior often results in poor aqueous dispersion and weak interfacial adhesion to hydrogel matrices.^[46,47] Therefore, maintaining the colloidal stability of LMNPs within hydrogel resins is crucial for achieving uniform dispersion and reliable VPP-based printing of LMNPs-integrated composites. So the well-dispersed pLMNPs were used in the experiments as the first step.

In the second step, pLMNPs were used together with different hydrogel systems. To assess the printability of pLMNPs-integrated resins, we systematically investigated three representative hydrogel systems—20 wt.% NiPAM, 10 wt.% HEA/5 wt.% PVA, and 10 wt.% GelMA, hereafter referred to as NiPAM, HEA/PVA, and GelMA, respectively. GelMA was synthesized via the reaction of gelatin with methacrylic anhydride, and its successful modification was confirmed by ¹H NMR, showing characteristic vinyl peaks at ≈ 5.2 and 5.6 ppm (Figure S4, Supporting Information). The preparation of pLMNPs-integrated hydrogel resins is detailed in Figure S5 (Supporting Information). pLMNPs were dispersed into three different prepolymer solutions—NiPAM, HEA/PVA, and GelMA—each containing a cross-linker, poly(ethylene glycol) diacrylate (PEGDA, average $M_n = 250 \text{ g mol}^{-1}$), and LAP photoinitiator. Colloidal stability was evaluated by monitoring the hydrodynamic size distributions of the pLMNPs over 4 h using DLS—a time frame sufficient to complete typical VPP-based printing tasks.

As shown in Figure 2a, the GelMA-pLMNPs exhibited an increased hydrodynamic size of 693.2 ± 12.9 nm compared to 215.6 ± 0.1 nm of pLMNPs sample in Milli-Q water. TEM images of pLMNPs within the 3D printing resins were obtained to assess morphological changes, revealing that all LMNPs maintained their spherical morphology and comparable size relative to pLMNPs (Figure S6a–c, Supporting Information). Therefore, the observed size increase is primarily attributed to the large molecular structure of GelMA and its coordination interactions with pLMNPs.^[48] Importantly, a stable monomodal size distribution was maintained over the 4 h duration, indicating good colloidal stability. Similarly, NiPAM- and HEA-based resins showed moderate size increases (312.3 ± 3.4 and 259.3 ± 4.2 nm, respectively) (Figure 2b,c; Figure S6d, Supporting Information), consistent with successful dispersion in polymeric media. Both systems preserved narrow and stable size distributions throughout the 4 h timeframe, sufficient for most 3D printing operations. Collectively, these results demonstrate the good colloidal stability of pLMNPs in all three hydrogel resin formulations, as evidenced by minimal size fluctuations and the absence of aggregation.

Viscosity is another key determinant of printability, influencing resin flow, layer recoating, and overall printing resolution. Rheological profiles of the three hydrogel resins, with and without pLMNPs, were assessed to evaluate their suitability for 3D

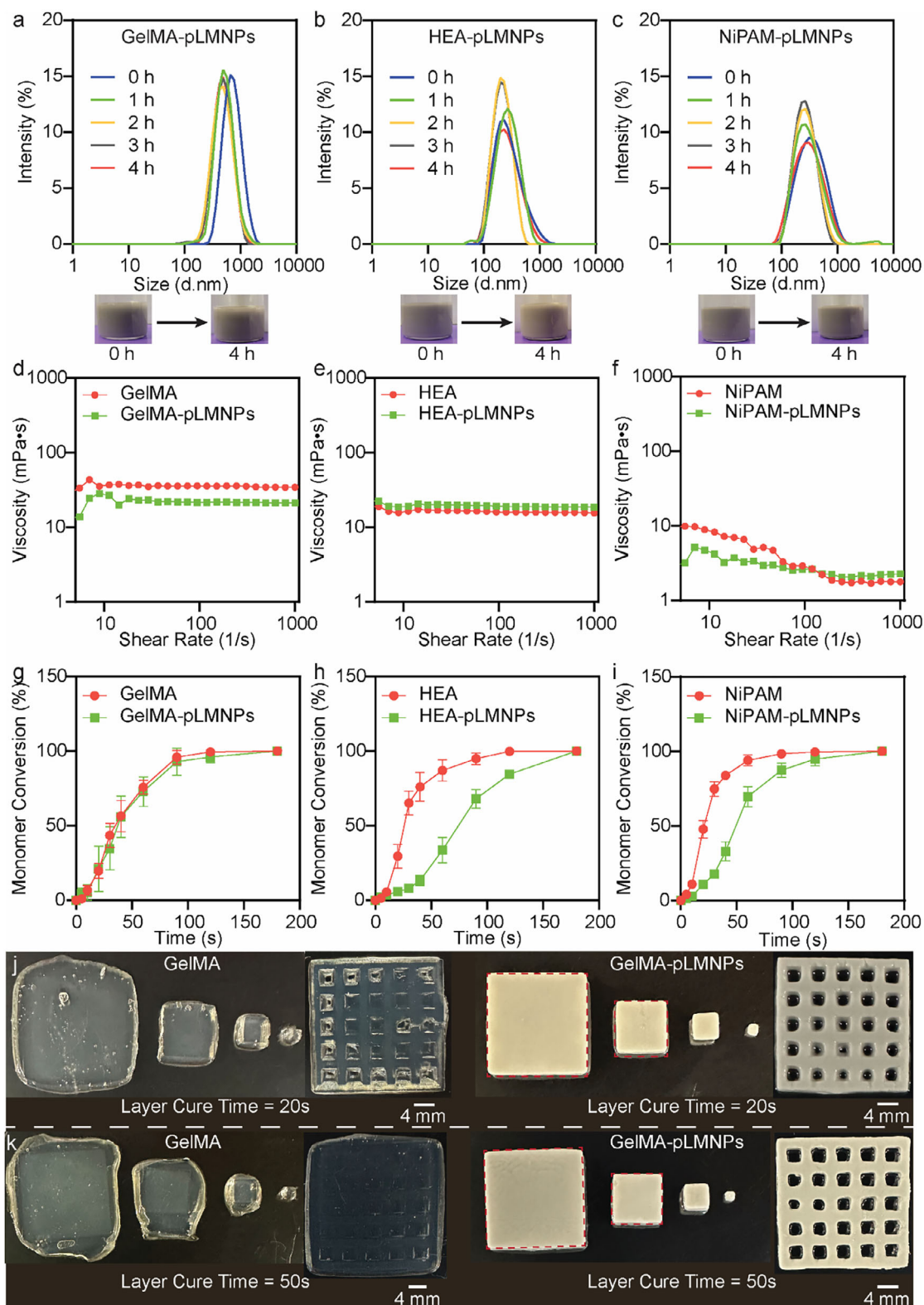


Figure 2. Printability of pLMNPs-integrated hydrogel resins. Size distribution and representative photographs of hydrogel resins containing pLMNPs: a) pLMNPs in GelMA; b) pLMNPs in HEA/PVA; c) pLMNPs in NiPAM; Viscosity measurements of corresponding formulations with and without pLMNPs: d) GelMA; e) HEA/PVA; and f) NiPAM; Photopolymerization kinetics of g) GelMA-based, h) HEA/PVA-based, and i) NiPAM-based hydrogel resins; Photographs of 3D-printed GelMA-based structures with varying pLMNPs content at different layer cure times: j) 20 s and k) 50 s, showing samples with 0 wt.% pLMNPs (left) and 1 wt.% pLMNPs (right). Corresponding CAD models are provided in Figures S7 and S8 (Supporting Information).

printing. As shown in Figure 2d, GelMA and GelMA-pLMNPs both exhibited shear-independent behavior, with stable viscosities across increasing shear rates. pLMNPs inclusion did not noticeably affect the viscosity. A similar pattern was observed in the HEA resin system (Figure 2e). In contrast, NiPAM-based resins displayed shear-thinning behavior (Figure 2f), which is generally favorable for extrusion or recoating processes. Again, pLMNPs had minimal influence on this behavior. All formulations maintained relatively low viscosities in the range of $\approx 1\text{--}50$ mPa-s, well below the maximum threshold (≈ 1300 mPa-s) for efficient VPP-based 3D printing.

To further assess the printability, the impact of pLMNPs on photopolymerization kinetics was investigated under violet light irradiation ($\lambda_{\text{max}} = 405$ nm, $I_0 = 1.5$ mW cm $^{-2}$) monitored in real time by FT-IR. As shown in Figures 2g–i, GelMA resins with and without pLMNPs showed similar polymerization kinetics. In contrast, slightly slower rates were observed in HEA and NiPAM systems upon pLMNPs incorporation, possibly due to pLMNPs modulating the propagation or initiation steps. Nevertheless, all formulations achieved over 80% monomer conversion within 120 s, confirming that pLMNPs integration does not hinder the photo-curing process essential for high-resolution 3D printing. Collectively, the well-maintained colloidal stability, favorable viscosity profiles, and efficient photopolymerization behavior established the opportunity for the excellent printability of pLMNPs-integrated hydrogel resins for VPP-based 3D printing applications.

2.3. Enhanced Printing Resolution and Expanded Processing Window using pLMNPs

To further elucidate the role of pLMNPs in improving the resolution and print fidelity of hydrogel constructs, we designed a series of rectangular and networked geometries to evaluate their printability and dimensional accuracy (CAD model Figures S7 and S8, Supporting Information). These structures were fabricated using a DLP 3D printer equipped with a digitally masked violet LED light source ($\lambda_{\text{max}} = 405$ nm, $I_0 = 3.5\text{--}4.0$ mW cm $^{-2}$). All prints were conducted with a fixed layer thickness of 0.05 mm and varied layer curing times (10–60 s). The actual printed lengths were compared against the CAD model length of 20.00 mm, as summarized in Tables S1–S3 (Supporting Information).

In GelMA hydrogels without pLMNPs, insufficient photopolymerization of the hydrogel at a 10 s curing time produced structurally compromised constructs, and the overall object length (19.11 ± 0.20 mm) remained within an acceptable range. Increasing the curing time to 20 s, or more, led to excessive hydrogel polymerization and material overgrowth, with dimensional deviations exceeding 20% at 50 s (Figure 2j–k; Table S1, Supporting Information). This overcuring resulted in blurred features and the accumulation of polymerized mass, compromising the resolution of printed structures.

In contrast, incorporating pLMNPs into the GelMA resin markedly improved print resolution. At a 20 s curing time, pLMNP-containing resins produced rectangular objects with sharp edges and high structural fidelity (Figure 2j). Even at extended curing times of up to 60 s, dimensional deviations remained below 10%, demonstrating an expanded processing

window. The as-printed structures maintained high geometric fidelity, with projected area variations below 20% relative to the CAD design. When networked hydrogels were printed, the pLMNP-free GelMA resins exhibited severe pore blockage and deformation of feature angles caused by overcuring. In contrast, the incorporation of pLMNPs enabled the fabrication of well-defined network structures with open, unobstructed pores at both 20 and 50 s layer curing times, achieving over 90% retention of projected area and approximately 90° rectangular features (Figure 2j–k), further validating the resolution-enhancing effect of pLMNPs inclusion.

A similar trend was observed in HEA-based resins, where rapid photopolymerization caused significant overcuring even at a 10 s curing time. Incorporation of pLMNPs moderated this effect, resulting in improved structural resolution across all tested exposure times (Figure S9 and Table S2, Supporting Information). NiPAM-based resins also benefited from pLMNPs integration, with enhanced edge definition and reduced dimensional error (Figure S9 and Table S3, Supporting Information). These improvements are attributed to the ability of pLMNPs to absorb the light, reduce light scattering over the entire UV–vis–NIR region, and enhance local light confinement within the transparent hydrogel matrix.^[49–51] This effect enables more precise photopolymerization, thereby improving resolution and structural control. Additionally, the presence of pLMNPs broadens the range of viable printing parameters, reducing the need for extensive optimization while enabling consistent fabrication of geometrically complex structures with high integrity.

2.4. 3D Printing of pLMNPs/Hydrogel Composites with Enhanced Structural and Mechanical Performance

On the basis of pLMNPs can reduce light scattering over the entire UV–vis–NIR region and enhance local light confinement within the transparent hydrogel matrix during 3D printing process, pLMNPs nanocomposites (made in two steps of initial coatings and secondary sonication process) were further employed in 3D printing to fabricate hybrid pLMNPs/hydrogel composites (LMHCs), demonstrating their excellent print fidelity and mechanical performance. These capabilities stem from the strong compatibility of pLMNPs with standard photocurable hydrogel resins.^[52,53] GelMA was selected as a representative natural hydrogel resin due to its excellent biocompatibility and broad use in biomedical applications.^[54]

Various geometries, including rectangular, triangular, and spherical LMHCs—were successfully 3D printed at a layer curing time of 30 s. All printed constructs displayed high dimensional accuracy relative to their CAD designs (target dimensions: 10.00 mm length/diameter, 3.00 mm thickness), with measured sizes of $10.17 \pm 0.30 \times 2.92 \pm 0.07$ mm (rectangle), $10.02 \pm 0.29 \times 2.94 \pm 0.08$ mm (triangle), and $10.60 \pm 0.38 \times 2.92 \pm 0.03$ mm (sphere) (Figure 3a). A more intricate butterfly-shaped structure was further fabricated, demonstrating outstanding structural integrity and high fidelity to the digital model (Figure 3b,c), with consistent agreement in overall dimensions as well as in feature angles, critical dimensions, and projected areas.

SEM imaging confirmed a uniform porous microstructure, indicating stable and homogeneous photopolymerization

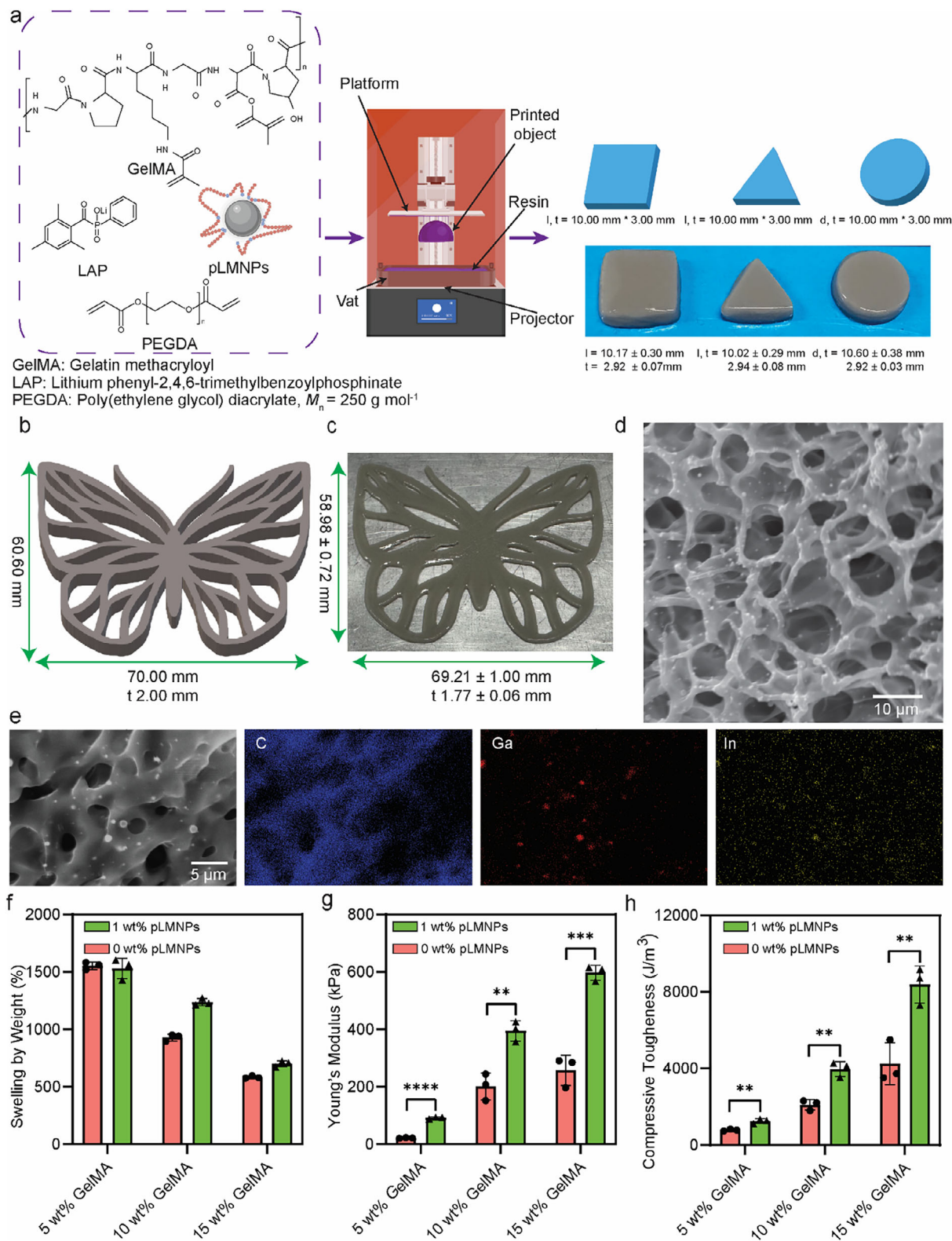


Figure 3. Fabrication and characterization of 3D-printed LMHCs. a) Schematic diagram of 3D-printed LMHCs, which are manufactured by using LAP as a photoinitiator, GelMA and PEGDA as crosslinker, and 1 wt.% pLMNPs. b) Computer-aided design (CAD) model of butterfly and c) the photograph of 3D printed butterfly LMHCs composed of 1 wt.% pLMNPs. d) SEM image of 3D-printed LMHCs with 1 wt.% pLMNPs. e) SEM image and EDS elemental mappings of LMHCs; the detected elements include C, Ga, and In. f) Swelling behaviors of 3D-printed GelMA hydrogels with/without pLMNPs. g) Young's modulus and h) Compressive toughness of 3D-printed GelMA hydrogels with/without pLMNPs.

throughout the printing process with no visible microstructural defects (Figure 3d). The presence of pLMNPs within the hydrogel network was also evident. To further confirm their integration, high-resolution SEM and EDS elemental mapping were conducted. As shown in Figure 3e, bright spots corresponding to pLMNPs were observed across the hydrogel surface and were confirmed by the co-localized Ga and In elemental signals. Concurrently, the carbon mapping validated the polymer network, further confirming successful incorporation of pLMNPs into the hydrogel matrix.

To further elucidate the toughening effect of pLMNPs in hydrogel matrices, we systematically investigated the mechanical performance of pLMNPs-reinforced GelMA hydrogels across various concentrations. Remarkably, even a low loading of 1 wt.% pLMNPs led to pronounced increases in both Young's modulus and compressive toughness across all tested GelMA formulations. For instance, at 15 wt.% GelMA, both the Young's modulus and compressive toughness were doubled from 257.37 ± 52.78 to 596.77 ± 26.11 kPa and from 4253.67 ± 1086.32 to 8396.00 ± 970.68 kPa, respectively, after pLMNPs incorporation (Figure 3f-g; Figure S10, Supporting Information). The reinforced mechanical properties were attributed to the formation of additional dynamic coordination bonds between Ga and the abundant carboxylic acid and amine groups on GelMA. These Ga-ligand interactions act as reversible dynamic covalent bonds, which can repeatedly dissociate and re-form under stress, thereby enhancing the toughness of the hydrogel. Moreover, the Ga ions serve as multifunctional cross-linking centers that improve interfacial adhesion and enable efficient stress transfer between polymer chains.^[47] These enhancements are particularly noteworthy in the context of soft material engineering, where achieving high strength and toughness without sacrificing printability or responsiveness remains a persistent challenge.

Our SEM and EDS mapping confirmed a uniform distribution of pLMNPs throughout the hydrogel matrix, with distinct Ga signals co-localized with the hydrogel network (Figure 3e). No discernible surface oxides were observed as the coating polymer on the LMNPs dynamically interplays with the solvent and Ga atoms exposed at the surface of LMNPs. As such, the interface of LMNPs could directly promote coordination bonds and non-covalent interactions with functional groups in GelMA, particularly amine and carboxylic acid moieties. These interactions could potentially introduce dynamic physical crosslinks within the network, enhancing energy dissipation during mechanical loading and thereby improving toughness. This mechanism is supported by prior works suggesting that metallic nanoparticles, especially those with electron-rich surfaces, can engage in strong interactions with polymer chains to create reinforced interfacial zones.^[55]

Notably, the incorporation of pLMNPs had minimal effect on the swelling behavior of the hydrogels, preserving their inherent flexibility and water content while enhancing their load-bearing capacity (Figure 3h). This unique combination of mechanical enhancement and structural fidelity makes pLMNPs-GelMA composites highly suitable for applications such as wearable devices, load-bearing scaffolds, and 4D-printed soft robots that demand both softness and durability.

From a broader perspective, LMNPs offer a dual-function strategy for hydrogel design: they both enhance mechanical

toughness and also improve photopolymerization control and printing resolution, as previously discussed. This synergistic effect sets pLMNPs-reinforced hydrogels apart from conventional nanocomposites that often face trade-offs between strength, swelling, and print fidelity. Thus, pLMNPs represent a promising class of multifunctional nanofillers that can drive the development of robust and precisely engineered soft materials.

2.5. Photothermal-Assisted Antibacterial Effects of 3D Printed Composites

Beyond their enhanced mechanical and structural properties, the incorporation of pLMNPs also endows the 3D-printed hydrogel composites with photothermal responsiveness,^[56,57] enabling potential antibacterial applications (Figure 4a). Under NIR irradiation (808 nm, 1 W cm^{-2}), the LMHCs (1 wt% pLMNPs and 15 wt% GelMA) exhibited a rapid and pronounced temperature increase, exceeding 55°C within 10 min (Figure 4b, inset). In contrast, pristine GelMA hydrogels, with no pLMNPs, showed negligible temperature changes, confirming that the photothermal response is solely attributed to pLMNPs.

To assess the antibacterial potential of LMHCs, we investigated their effect on two representative bacterial strains: *Escherichia coli* (*E. coli*, Gram-negative) and *Staphylococcus aureus* (*S. aureus*, Gram-positive). Bacteria were co-cultured with either hydrogels or LMHCs for 2 h before exposure to NIR irradiation (1 W cm^{-2} for 10 min), followed by incubation for an additional 12 h. Colony counting assays revealed that LMHCs under NIR treatment resulted in almost complete bacterial inhibition (Figure 4c,d; Figure S11, Supporting Information). Specifically, LMHCs with NIR irradiation achieved $89.9 \pm 16.6\%$ inhibition for *E. coli* and $99.5 \pm 0.8\%$ inhibition for *S. aureus*, whereas all other groups—including hydrogels with or without NIR and LMHCs without NIR—showed high bacterial viability (Figure 4e,f).

These results confirm that the antibacterial performance of LMHCs arises primarily from photothermal heating, as non-heated controls showed no significant inhibition, in line with the absence of temperature elevation. While previous studies have attributed the antibacterial effects of LMNPs to the release of Ga^{3+} ions,^[58] such effects appear to be minimized in our system. Here, the hydrogel matrix hinders ion release either by encapsulation of pLMNPs or by limiting Ga^{3+} to sub-therapeutic concentrations.^[59] Therefore, in the short-term treatment window assessed, photothermal action is the dominant antibacterial mechanism.

Further validation was obtained using live/dead bacterial staining and confocal microscopy (Figure 5a,b). In both *E. coli* and *S. aureus*, only LMHCs under NIR treatment exhibited significant red fluorescence, indicating high bacterial mortality, whereas all other groups showed predominantly green signals, reflecting live cells. These findings demonstrate that the photothermal effect of LMHCs is effective across both Gram-negative and -positive bacteria, highlighting their broad-spectrum potential.

SEM imaging of bacteria post-treatment further confirmed these observations (Figure 5c,d). Bacterial cells treated with NIR-irradiated LMHCs exhibited clear membrane damage and shrinkage, consistent with thermal disruption. In contrast, cells

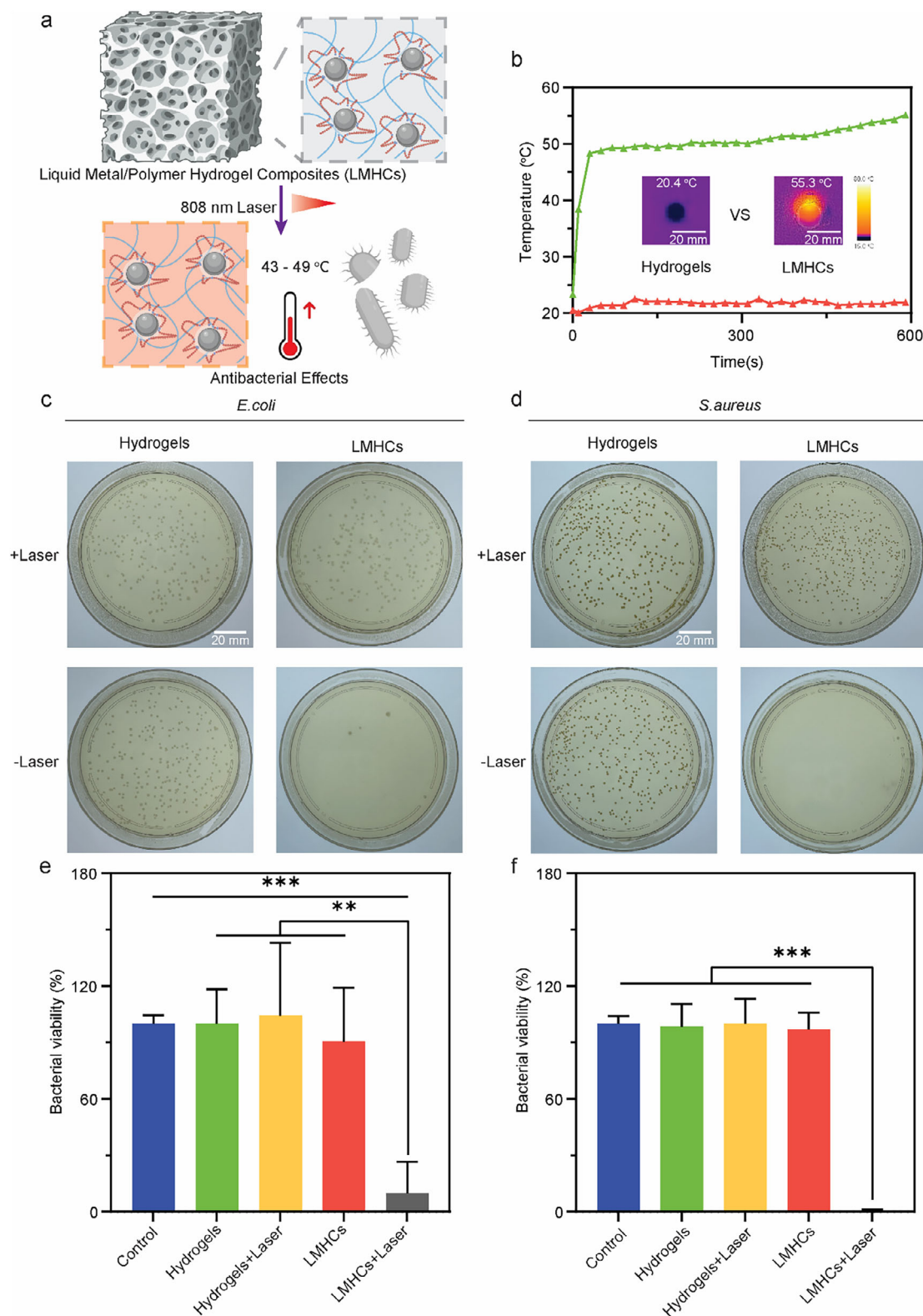


Figure 4. a) Diagrammatic illustration of antibacterial performance of LMHCs. b) Photothermal effects of LMHCs and H₂O under 1 W cm⁻² 808 nm NIR laser. c) Photographs of the *E. coli* bacterial colony incubated with hydrogels, hydrogels+Laser, LMHCs, and LMHCs+Laser. d) Photographs of the *S. aureus* bacterial colony incubated with hydrogels, hydrogels+Laser, LMHCs, and LMHCs+Laser. Quantitative analysis of *E. coli* e) and *S. aureus* f) co-cultured with different samples.

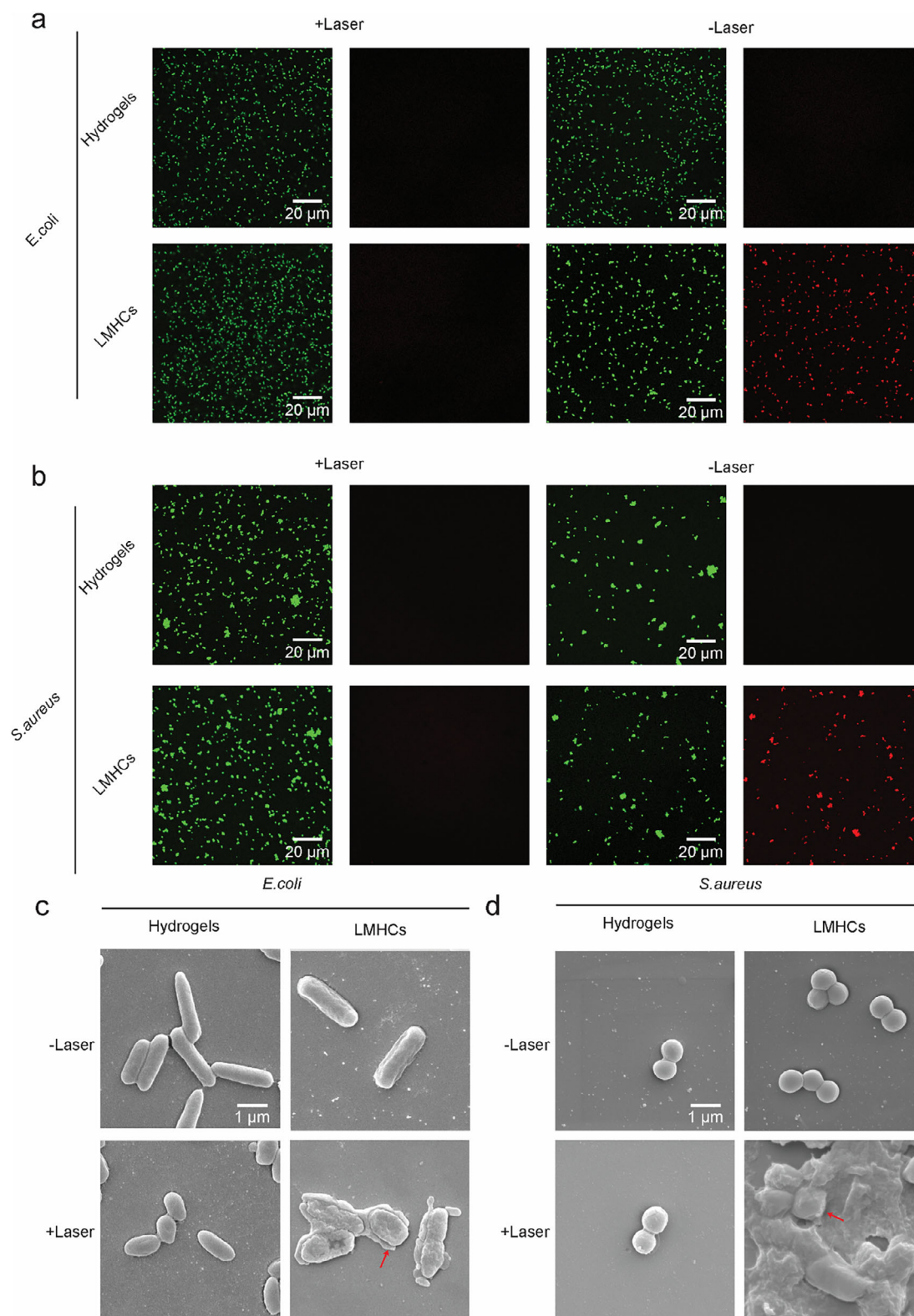


Figure 5. Confocal microscopy images of *E. coli* a) and *S. aureus* b) treated by hydrogels and LMHCs with/without laser treatment. The bacteria were stained with green fluorescence signals, indicating live cells, and red signals indicating dead cells. The scale bar is 20 μm . SEM images of *E. coli* c) and *S. aureus* d) treated by hydrogels and LMHCs with/without laser treatment.

in the other groups remained structurally intact, displaying normal morphology and membrane integrity. In order to ensure the feasibility of their future biomedical applications, the cytotoxicity of the hydrogels and LMHCs was further assessed using human embryonic kidney (HEK-293) cells. Both hydrogel- and LMHC-treated cells showed over 90% viability (Figure S12, Supporting Information), indicating excellent biocompatibility.

3. Discussion

This work presents a novel strategy to fabricate high-resolution 3D printed hybrid hydrogels by integrating pLMNPs into a GelMA-based matrix, resulting in LMHCs with significantly improved printability, mechanical properties, and photothermal-assisted antibacterial activity. A major challenge in VPP-based printing hydrogels lies in balancing print fidelity with material performance, especially for applications requiring precise architecture such as soft tissue scaffolds, microrobots, or bioelectronics. In this context, the key advancement of our work is the markedly improved resolution and structural integrity of 3D printed constructs enabled by pLMNPs integration. Beyond serving as functional nanofillers that enhance the mechanical properties of printed hydrogels (increasing Young's modulus from 257.37 ± 52.78 to 596.77 ± 26.11 kPa, and compressive toughness from 4253.67 ± 1086.32 to 8396.00 ± 970.68 kPa), the pLMNPs also modulate the photopolymerization kinetics of the hydrogel resin, enabling higher-resolution printing with fewer artifacts and reduced overcuring. This improvement is evident in our ability to fabricate intricate and geometrically complex designs—including rectangles, spheres, and butterfly patterns—with high fidelity to the CAD models. Such enhanced printability expands the design space for customized and application-specific hydrogel devices, while importantly maintaining the processing ease of the parent hydrogel system.

Further demonstration indicates that LMHCs exhibit potent photothermal responsiveness under 808 nm NIR irradiation, enabling effective on-demand antibacterial activity against both *E. coli* and *S. aureus*. In contrast, pristine hydrogels and LMHCs without NIR showed minimal bactericidal activity, confirming the photothermal origin of the effect. These results underscore the critical interplay between 3D-printed structural fidelity and photothermal functionality in LMHCs. The high-resolution architectures enable homogeneous LMNP dispersion, facilitating spatially uniform and controllable heat generation under NIR stimulation. This structural–functional synergy provides a powerful design strategy for developing next-generation biofunctional hydrogels.

In summary, this work establishes LMHCs as a versatile class of engineerable soft matter characterized by high structural precision, tunable mechanical performance, and on-demand antibacterial functionality, thereby bridging material design and biomedical applications. The synergistic effects derived from pLMNPs inclusion not only optimize the photopolymerization behavior and mechanical properties of the as-printed hydrogels but also enable dynamic responsiveness under external stimuli. These combined features open new avenues for developing multifunctional and stimuli-responsive hydrogels, holding great promise for diverse biomedical and technological applications such as

tissue regeneration, infection prevention, and intelligent soft robotic systems.

4. Experimental Section

Materials: Gelatin (from bovine skin), methacrylic anhydride (94%), PEGDA ($M_n = 250$ g mol⁻¹), gallium-indium eutectic (EGaIn, Ga 75.5% and In 24.5%, $\geq 99.99\%$), N-isopropylacrylamide, 2-hydroxyethyl acrylate, LAP, DPBS, and LB agar were purchased from Sigma–Aldrich. The monomers were passed through an alumina column (basic, 100–200 mesh, purchased from Fluka) to remove inhibitors prior to use. All other reagents were used as received unless otherwise specified.

Fabrication and Characterization of pLMNPs: The RAFT polymer was prepared as stated in the previous literature.^[29] EGaIn (100 mg) and RAFT polymer (200 mg) were mixed in 10 mL Milli-Q water, then placed in an ice bath for 30 min. Afterward, the EGaIn and RAFT polymer solution was sonicated using Sonics VCX-750 Vibra Cell Ultra Sonic Processor set at 30% power, equipped with a 6 mm sonication probe, for 30 min. The pLMNPs were collected after ultra-centrifugation at 15 000 rpm for 10 min.

TEM and EDS Mapping: The morphology and element composition of pLMNPs were analyzed on a Hitachi HF5000 Cs-STEM/TEM system equipped with Oxford EDX 2 × 1sr 100 mm SDD EDX detectors and Gatan US1000 post spectrometer camera. Ga, In, O, and S signals from EDS analysis were collected to demonstrate the element distribution within pLMNPs.

DLS: The hydrodynamic size and zeta potential of Bare LMNPs, RAFT polymer and pLMNPs were monitored by Malvern Zetasizer Ultra equipped with a solid-state He-Ne laser, $\lambda = 633$ nm.

FT-IR: FT-IR spectra were recorded on a Nicolet 6700 spectrometer under attenuated total reflectance (ATR). The spectra were collected over 512 scans with a spectral resolution of 2 cm⁻¹. The monomer conversions were also monitored by FT-IR over 32 scans with a spectral resolution of 2 cm⁻¹.

Photothermal Evaluation of pLMNPs: The photothermal effect of pLMNPs (1 mg mL⁻¹) under 808 nm laser was investigated at different laser intensities of 0, 0.3, 0.5, 1.0, and 2.0 W cm⁻² for 10 min. Meanwhile, 0.2, 0.5, 1.0, and 2.0 mg mL⁻¹ of pLMNPs were exposed under 1 W cm⁻² 808 nm laser. For photothermal stability during ten photothermal heating and cooling cycles, the pLMNPs solutions were exposed to 1 W cm⁻² 808 nm laser for 3 min and cooling for 6 min. The temperatures were recorded at intervals of 10 s using an infrared thermal camera FLIR ONE PRO.

Synthesis of GelMA: GelMA was synthesized following established protocols with slight adjustment.^[60] Gelatin (10 g, 3.18 mmol of free amino groups) was dissolved in 100 mL of DPBS under continuous stirring at 60 °C. Subsequently, 0.6 mL of methacrylic anhydride (4.03 mmol) was added dropwise to the gelatin solution, and the reaction mixture was stirred for 5 h at 60 °C. The reaction solution was quenched by an equal volume of DPBS followed by centrifuging at 3000 rpm for 5 min to remove any precipitates. The quenched solution was then dialyzed using Snake-Skin 3.5K MWCO dialysis tubing in Milli-Q water, with daily water replacement, over the course of 1 week. Finally, the GelMA product was obtained through freeze-drying of the dialyzed GelMA solution.

NMR: ¹H NMR spectra analysis was conducted using a Bruker Avance 400 MHz spectrometer at 298 K.

3D Printing of pLMNPs-Integrated Hydrogels: Three 3D printing resins were first prepared. 20% NiPAM contains 2.0 g NiPAM (20 wt.%), 0.5 g PEGDA (5 wt.%), and 0.05 g LAP (0.05 wt.%) in 10 mL Milli-Q water; 10% HEA/5% PVA contains 1.0 g HEA (10 wt.%), 0.5 g PVA (5 wt.%), 0.5 g PEGDA (5 wt.%), and 0.05 g LAP (0.05 wt.%) in 10 mL Milli-Q water; 10% GelMA contains 1.0 g GelMA (10 wt.%), 0.5 g PEGDA (5 wt.%), and 0.05 g LAP (0.05 wt.%) in 10 mL Milli-Q water. 0.1 g LMNPs were then incorporated into these three resins to obtain the LMNPs-integrated hydrogel resins. The 3D printing resins were sonicated for 20 min, then 3D printing resins were subjected to Anycubic 3D Printer (stereolithography-based 3D printer, $\lambda_{\text{max}} = 405$ nm, $I_0 = 3,500 - 4,000$ $\mu\text{W cm}^{-2}$). 3D CAD design models and .stl files were created using Autodesk Fusion 360. The regular cure time (10–60 s) per layer and the first 5 (bottom) layers of objects were

irradiated for 15–65 s as stated in the main text applies in the process of 3D printing. The Z lift distance was 1 mm, the Z lift speed was 3 mm s^{−1}, and the Z retract speed was 3 mm s^{−1}.

SEM: SEM images were obtained by a JEOL JSM-7100F field emission scanning electron microscope with a hot (Schottky) electron gun. SEM images and EDS mapping were collected by HITACHI SU3500 SEM-EDS equipped with an Oxford Xmax SDD EDS detector.

Compression Tests: Compression tests were conducted on a Mark-10 test stand to study the mechanical properties of 3D-printed specimens. Cylindrical samples with a diameter of 10 mm and a height of 3 mm were 3D printed with the procedure described above. The rate of displacement was set as 1.1 mm min^{−1} for all tests. The dimensions of printed samples were measured using an electronic digital caliper. Young's modulus is obtained using the slope value of the linear part of the stress-strain curve. The compressive toughness (U_t) was calculated as

$$U_t = \int_0^{\sigma_{\max}} \sigma d\epsilon \quad (1)$$

where σ was stress in the stress–strain curves, ϵ was defined as strain, and $\sigma_{\max}$ was the maximum stress during the compression.

Swelling Ratio: 3D printed hydrogels were freeze-dried and weighed as initial swelling weight (W_{dry}). The hydrogels were then subjected to Milli-Q water for a certain period of swelling. The weights of swollen hydrogels (W_{wet}) were recorded over the course of swelling. The swelling ratio was determined by $W_{\text{wet}}/W_{\text{dry}}$ from three parallel samples.

Bacterial Culture: *E. coli* and *S. aureus* were obtained from the American Type Culture Collection. These two species were chosen as representatives of Gram-negative and -positive bacteria, respectively, accounting for the most commonly occurring infective pathogens. These species were cultured on LB agar at 37 °C for 24 h. A single colony of bacteria was isolated and inoculated in LB broth media for 8–10 h in a shaking incubator (220 rpm) at 37 °C. The concentration of bacteria was monitored by measuring the optical density at 600 nm (OD_{600}) by the EnSight Multi-mode Plate Reader.

In Vitro Antibacterial Experiments: The bacteria were diluted to the appropriate level ($OD_{600} = 0.1$) and then treated with GelMA hydrogels or LMHCs for 2 h at 37 °C on a shaking incubator (200 rpm). The groups of Hydrogels-NIR and LMHCs-NIR were separately irradiated with an 808 nm laser at a power density of 1.0 W cm^{−2} for 10 min. Then, the solution of bacteria was serially diluted and spread on LB agar plates. The colonies were counted (Image J) to determine the bacterial effects of different conditions after cultivation for 12 h at 37 °C ($n = 3$).

Morphology Observations of Bacteria: For SEM observations, the bacterial cells were fixed overnight onto glass coverslips with 4% paraformaldehyde and 2% glutaraldehyde in a 50:50 ratio. After washing with PBS three times, the bacterial cells were dehydrated through sequential treatments of 30%, 50%, 75%, and 100% ethanol for 1 h. The samples were finally dried overnight in hexamethyldisilazane solution and subjected to a 15 nm thick platinum coating to improve the conductivity for SEM (JEOL JSM-7100F) observations.

Live/Dead Staining Assay: The treated bacteria were stained using LIVE/DEAD BacLight Bacterial Viability Kit to assess the antibacterial performance according to the supplier's manual for 20 min in the dark. The bacterial cells were washed with PBS three times after staining and subjected to a clean round glass slide with a diameter of 0.5 cm. The Live/Dead images of bacteria were observed under a Zeiss LSM 710 confocal laser scanning microscope.

Biocompatibility Test: Human embryonic kidney (HEK-293) cells were obtained from ATCC (USA). Hydrogels and LMHCs were incubated in complete culture medium for 24 h, after which the supernatant was collected as the hydrogel and LMHC extracts. HEK-293 cells were seeded in 24-well plates at a density of 50 000 cells per well and cultured for 24 h. The medium was then replaced with the corresponding hydrogel or LMHC extract and incubated for an additional 24 h. Subsequently, 50 μ L of alamarBlue HS Cell viability reagent was added to each well. After 3 h of incubation, absorbance was measured at 570 and 600 nm using an EnSight plate reader.

Statistical Analysis: All the results were repeated with at least three identical samples. All data are expressed as means $\pm$ standard deviation (SD) with statistical significance assessed by unpaired two-sided *t*-tests. GraphPad Prism (v10.0.2) was used for data analysis.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

Acknowledgements

The authors would like to acknowledge the support from the National Health and Medical Research Council (APP1196850 and 1197373), Australian Research Council Discovery Projects (DP240102315 and DP240101086) and Queensland-Chinese Academy of Sciences (Q-CAS) Collaborative Science Fund (QCAS2022016). This research was supported by the Queensland Government's Advance Queensland Industry Research Fellowships (AQIRF). This work used the Queensland node of the NCRIS-enabled Australian National Fabrication Facility (ANFF), and the Centre for Microscopy and Microanalysis (CMM).

Conflict of Interest

The authors declare no conflict of interest.

Author Contributions

X.H. and L.Z. contributed equally to this work. R.Q., K.K.Z., L.Z. and T.P.D. conceived the idea and supervised this project. X.H. and L.Z. designed, performed the experiments, and analyzed the results. X.H. conducted polymers, nanoparticles, SEM and EDS mappings. N.K.A.N. prepared GelMA, 3D-printed objects and performed mechanical tests. L.L. and X.H. conducted confocal microscopy imaging. X.G. and X.H. performed the bacterial experiments. X.H., L.Z., and R.Q. wrote the paper with input from all authors. L.Z., R.Q., Y.L., K.K.Z. and T.P.D. provided critical feedback and helped revise this article.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

3D Printing, antibacterial materials, hydrogels, liquid metal nanoparticles, vat Photopolymerization

Received: September 7, 2025
Revised: October 31, 2025
Published online: November 20, 2025

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